

Task 1: Data Modeling & Cleaning

- Load and transform datasets in Power Query

	A _C Transaction_ID	A _C Customer_ID	A _C Store_ID	A _C Product_Category	Transaction_Date	L2 Amount	A _C Promotion_Applied	i ₃ Year	A _C Month Name
1	T0531	C001	S20	Apparel	29-10-2024	335.04	Yes	2024	October
2	T0597	C001	S09	Groceries	11-06-2024	551.76	Yes	2024	June
3	T0454	C001	S12	Home & Living	16-03-2023	527.9	No	2023	March
4	T0857	C001	S16	Apparel	20-06-2024	545.13	No	2024	June
5	T0550	C001	S02	Electronics	02-07-2024	765.62	Yes	2024	July
6	T0818	C001	S14	Electronics	23-10-2024	632.38	No	2024	October
7	T0274	C001	S19	Home & Living	08-12-2023	461.93	Yes	2023	December
8	T0356	C002	S18	Electronics	24-03-2024	236.76	Yes	2024	March
9	T0836	C002	S11	Groceries	18-02-2024	362.58	No	2024	February
10	T0580	C002	S02	Apparel	27-05-2023	262.28	Yes	2023	May
11	T0800	C002	S07	Apparel	14-03-2023	361.94	No	2023	March
12	T0024	C002	S14	Groceries	25-03-2024	103.15	Yes	2024	March
13	T0805	C003	S17	Groceries	26-10-2023	213.1	Yes	2023	October
14	T0797	C003	S01	Apparel	05-01-2024	599.4	Yes	2024	January
15	T0776	C004	S06	Groceries	23-07-2024	108.38	No	2024	July
16	T0506	C004	S20	Apparel	21-08-2023	329.67	Yes	2023	August
17	T0530	C005	S15	Home & Living	21-10-2023	679.57	No	2023	October
18	T0344	C005	S16	Groceries	29-05-2023	608.93	No	2023	May
19	T0355	C005	S13	Apparel	06-08-2024	212.41	No	2024	August
20	T0023	C005	S09	Groceries	31-12-2023	140.64	No	2023	December
21	T0758	C005	S09	Electronics	14-08-2024	122.74	No	2024	August
22	T0798	C006	S02	Home & Living	21-05-2024	85.76	No	2024	May
23	T0025	C006	S07	Apparel	21-06-2024	50.75	No	2024	June
24	T0651	C006	S03	Apparel	16-05-2024	33.22	No	2024	May
25	T0243	C006	S01	Electronics	06-06-2024	200.92	Yes	2024	June
26	T0631	C006	S15	Home & Living	06-07-2024	391.25	Yes	2024	July
27	T0202	C006	S01	Apparel	07-08-2024	90.58	Yes	2024	August
28	T0710	C007	S09	Electronics	02-08-2024	473.67	No	2024	August
29	T0210	C007	S05	Groceries	11-05-2024	366.65	Yes	2024	May
30	T0446	C007	S07	Groceries	20-04-2024	689.38	No	2024	April
31	T0883	C007	S10	Apparel	08-08-2023	324.79	Yes	2023	August
32	T0122	C007	S14	Groceries	11-07-2024	380.44	Yes	2024	July
33	T0669	C007	S10	Apparel	16-02-2024	521.4	No	2024	February
34	T0267	C007	S03	Home & Living	31-03-2024	609.11	Yes	2024	March
35	T0743	C008	S04	Apparel	27-02-2024	380.14	No	2024	February
36	T0934	C008	S07	Apparel	12-02-2023	373.79	No	2023	February
37	T0499	C008	S20	Apparel	09-10-2024	206.31	No	2024	October
38	T0078	C009	S10	Groceries	04-05-2023	310.7	Yes	2023	May

- Handle duplicates, missing values, and ensure correct data types

	A _C Customer_ID	i ₃ Age	A _C Gender	A _C Region	A _C Income_Level	Membership_Since	A _C Preferred_Channel	i ₃ Membership_Duration
1	C001		58 Female	North	Low	18-08-2023	Online	853
2	C002		48 Male	North	High	17-09-2019	Store	2284
3	C003		34 Female	East	Medium	06-03-2023	Store	1018
4	C004		27 Female	Central	Low	30-06-2021	Store	1632
5	C005		40 Male	South	High	12-09-2022	Online	1193
6	C006		58 Male	South	High	30-10-2018	Online	2606
7	C007		38 Female	East	High	01-10-2019	Store	2270
8	C008		42 Female	South	Low	25-02-2018	Online	2853
9	C009		30 Male	North	Medium	27-12-2019	Online	2183
10	C010		30 Female	Central	Medium	10-06-2019	Online	2383
11	C011		43 Male	West	Medium	19-04-2019	Store	2435
12	C012		55 Female	South	High	28-01-2018	Store	2881
13	C013		59 Male	North	Medium	27-10-2023	Online	783
14	C014		43 Male	West	High	31-12-2018	Store	2544
15	C015		22 Male	Central	High	10-06-2019	Online	2383
16	C016		41 Female	West	High	28-08-2021	Online	1573
17	C017		21 Female	North	Medium	01-06-2020	Online	2026
18	C018		43 Male	West	Low	27-10-2020	Online	1878
19	C019		49 Female	East	Medium	23-10-2020	Store	1882
20	C020		57 Male	West	High	05-07-2022	Store	1262
21	C021		21 Male	South	Low	07-07-2020	Store	1990
22	C022		40 Female	South	Medium	07-08-2024	Online	498
23	C023		52 Female	East	Low	10-11-2022	Online	1134
24	C024		31 Male	North	Medium	12-03-2022	Online	1377
25	C025		41 Female	South	High	22-03-2023	Store	1002
26	C026		44 Female	Central	High	30-04-2021	Online	1693
27	C027		46 Male	South	Low	13-07-2022	Store	1254
28	C028		47 Male	South	High	30-04-2019	Store	2424

- Create calculated columns:
 - o Membership_Duration = Today – Membership_Since

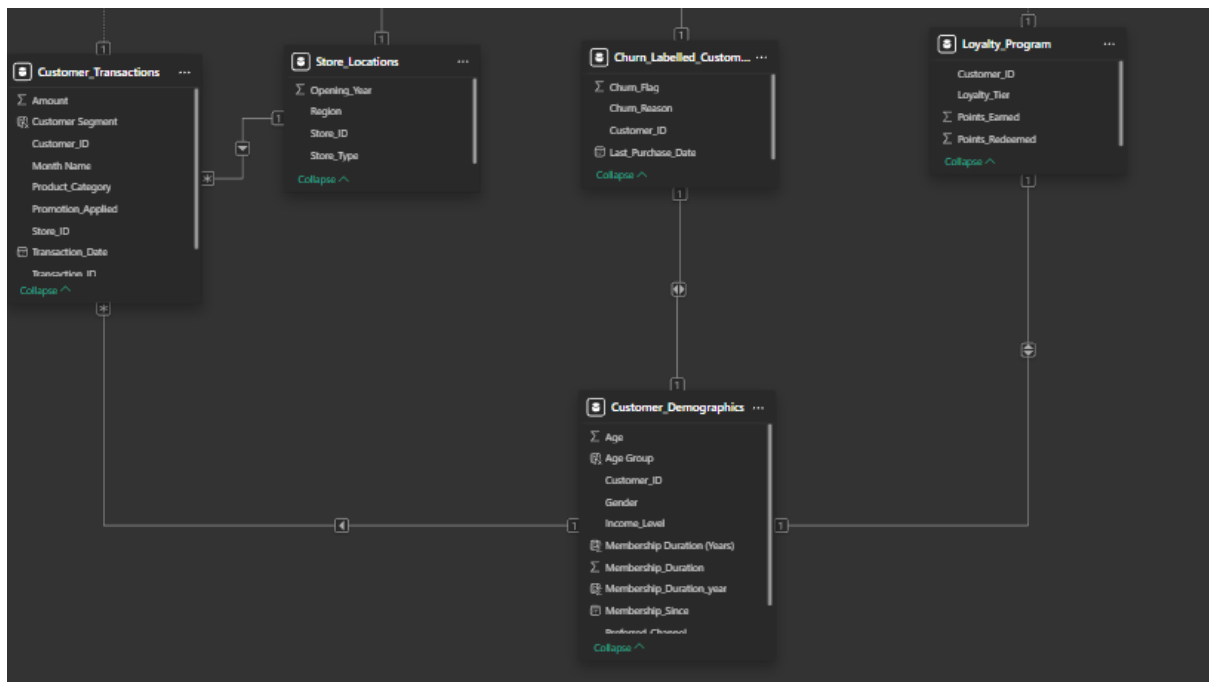
Membership_Since	Preferred_Channel	Membership_Duration
18-08-2023	Online	853
17-09-2019	Store	2284
06-03-2023	Store	1018
30-06-2021	Store	1632
12-09-2022	Online	1193
30-10-2018	Online	2606
01-10-2019	Store	2270
25-02-2018	Online	2853
27-12-2019	Online	2183
10-06-2019	Online	2383
19-04-2019	Store	2435
28-01-2018	Store	2881
27-10-2023	Online	783
31-12-2018	Store	2544
10-06-2019	Online	2383
28-08-2021	Online	1573
01-06-2020	Online	2026
27-10-2020	Online	1878
23-10-2020	Store	1882
05-07-2022	Store	1262
07-07-2020	Store	1990
07-08-2024	Online	498
10-11-2022	Online	1134
12-03-2022	Online	1377
22-03-2023	Store	1002
30-04-2021	Online	1693
13-07-2022	Store	1254
30-04-2019	Store	2424
07-09-2023	Store	833
15-12-2019	Store	2195
28-11-2020	Store	1846
31-05-2023	Store	932
20-07-2021	Online	1612
08-06-2021	Online	1654
04-11-2023	Store	775
06-05-2018	Store	2783
21-06-2019	Online	2372
02-01-2024	Store	716

- o Extract Transaction_Year, Transaction_Month

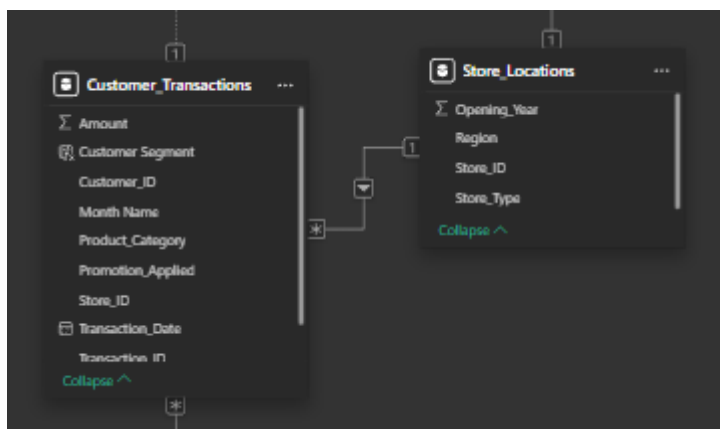
Product_Category	Transaction_Date	Year	Month Name
Apparel	20-06-2024	2024	June
Home & Living	16-03-2023	2023	March
Electronics	23-10-2024	2024	October
Electronics	02-07-2024	2024	July
Apparel	19-10-2024	2024	October
Groceries	11-06-2024	2024	June
Home & Living	08-12-2023	2023	December
Electronics	24-03-2024	2024	March
Groceries	18-02-2024	2024	February
Apparel	27-05-2023	2023	May
Apparel	14-03-2023	2023	March
Groceries	25-03-2024	2024	March
Groceries	26-10-2023	2023	October
Apparel	05-01-2024	2024	January
Apparel	21-08-2023	2023	August
Groceries	23-07-2024	2024	July
Home & Living	21-10-2023	2023	October
Electronics	14-08-2024	2024	August
Apparel	06-08-2024	2024	August
Groceries	29-05-2023	2023	May
Groceries	31-12-2023	2023	December
Home & Living	06-07-2024	2024	July
Home & Living	21-05-2024	2024	May
Electronics	06-06-2024	2024	June
Apparel	07-08-2024	2024	August
Apparel	16-05-2024	2024	May
Apparel	21-06-2024	2024	June
Apparel	08-08-2023	2023	August
Groceries	11-05-2024	2024	May
Apparel	16-02-2024	2024	February
Home & Living	31-03-2024	2024	March
Groceries	11-07-2024	2024	July
Groceries	20-04-2024	2024	April
Electronics	02-08-2024	2024	August
Apparel	27-02-2024	2024	February

- Create a basic Data Model view

- One-to-Many: Customer_Demographics → Transactions, Loyalty_Program, Churn_Labelled_Customers



- Many-to-One: Transactions → Store_Locations



Task 2: Churn & Retention Metrics

- Create Churn Rate KPI = (Churned Customers / Total Customers) * 100

24.50
Churn Rate (%)

Churn Rate (%) =

```
DIVIDE(  
  [Churned Customers],  
  [Total Customers],  
  0  
) * 100
```

- Visualize churn rate by:



● Funnel Chart: Total Customers → Repeat Customers → Churned



Task 3: Repeat Purchase Analysis

- Segment customers:

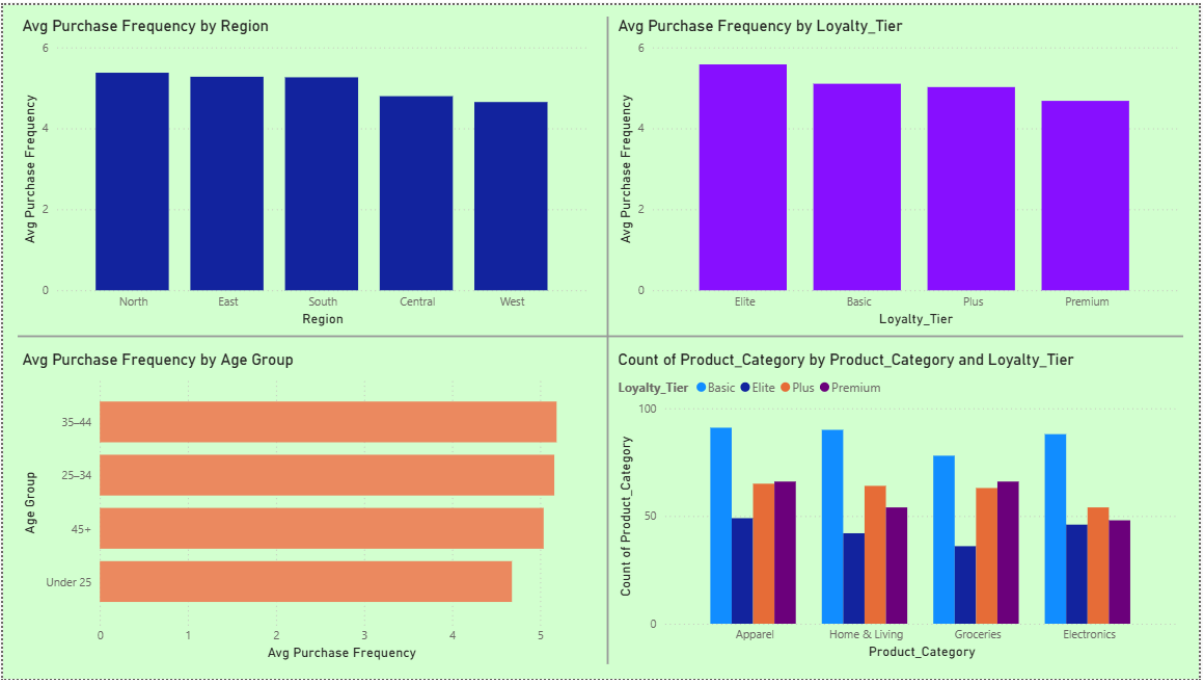
Customer_ID	Count of Transaction_ID	Customer Segment (Measure)
C014	10	High-Tier
C049	10	High-Tier
C099	10	High-Tier
C132	10	High-Tier
C160	10	High-Tier
C009	9	High-Tier
C013	9	High-Tier
C053	9	High-Tier
C072	9	High-Tier
C081	9	High-Tier
C129	9	High-Tier
C137	9	High-Tier
C149	9	High-Tier
C176	9	High-Tier
C180	9	High-Tier
C182	9	High-Tier
C196	9	High-Tier
C026	8	Mid-Tier
C047	8	Mid-Tier
C087	8	Mid-Tier
C166	8	Mid-Tier
C197	8	Mid-Tier
C001	7	Mid-Tier
C007	7	Mid-Tier
C015	7	Mid-Tier
C022	7	Mid-Tier
C023	7	Mid-Tier

```

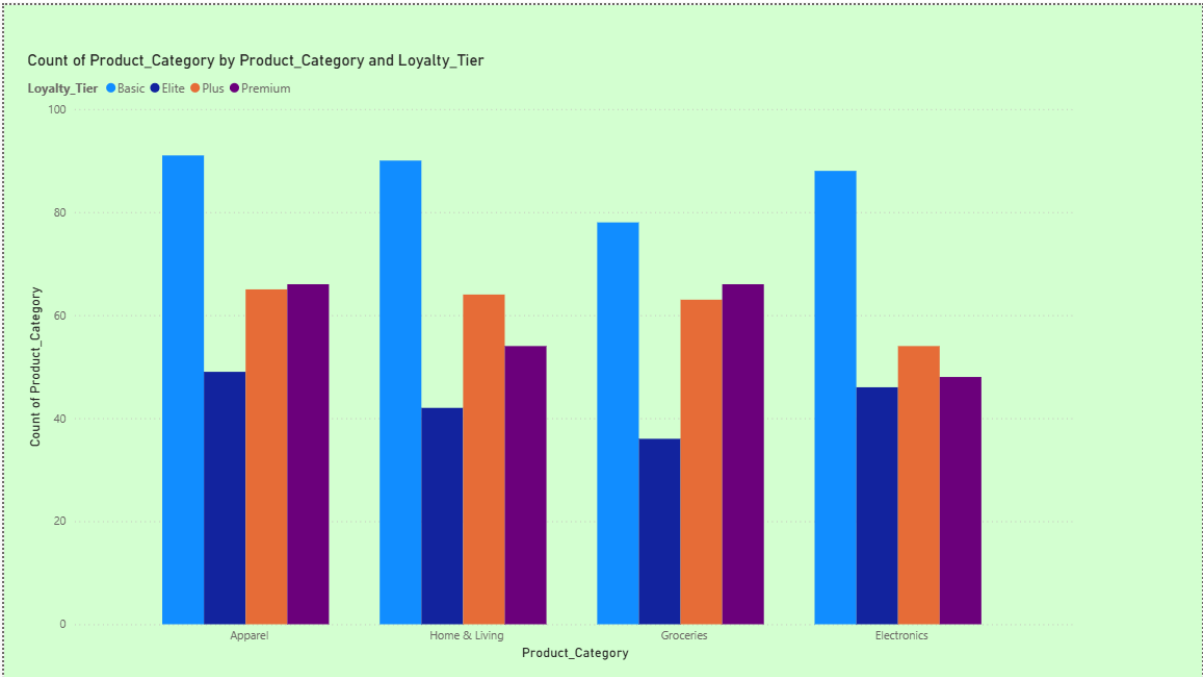
Customer Segment (Measure) =
VAR PurchaseCount =
    COUNT(Customer_Transactions[Transaction_ID])
RETURN
SWITCH(
    TRUE(),
    PurchaseCount <= 3, "Low-Tier",
    PurchaseCount <= 8, "Mid-Tier",
    "High-Tier"
)

```

- Compare avg. purchase frequency by Region, Age Group, Loyalty Tier

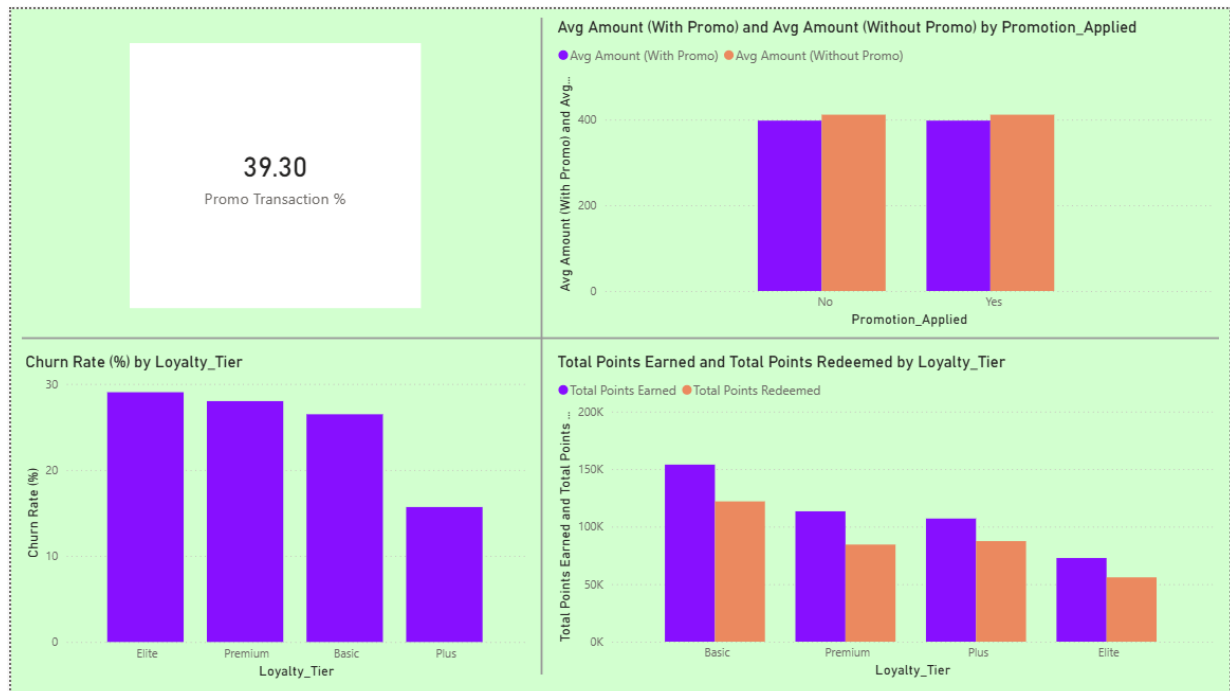


- Identify most purchased product categories by loyal customers



Task 4: Promotion & Loyalty Impact

- Calculate % of transactions with promotion applied.
- Compare average purchase amount with vs. without promotions.
- Analyze churn rate across loyalty tiers.
- Visualize Points Earned vs. Redeemed by Tier (Clustered Column Chart).



```
Promo Transaction % =
DIVIDE(
    [Promo Transactions],
    [Total Transactions],
    0
) * 100
```

```
Churn Rate (%) =
DIVIDE(
    [Churned Customers],
    [Total Customers],
    0
) * 100
```

```
Avg Amount (Without Promo) =  
CALCULATE(  
    [Avg Purchase Amount],  
    Customer_Transactions[Promotion_Applied] = "No"  
)
```

```
Avg Amount (With Promo) =  
CALCULATE(  
    [Avg Purchase Amount],  
    Customer_Transactions[Promotion_Applied] = "Yes"  
)
```

```
Total Points Earned =  
SUM(Loyalty_Program[Points_Earned])
```

```
Total Points Redeemed =  
SUM(Loyalty_Program[Points_Redeemed])
```

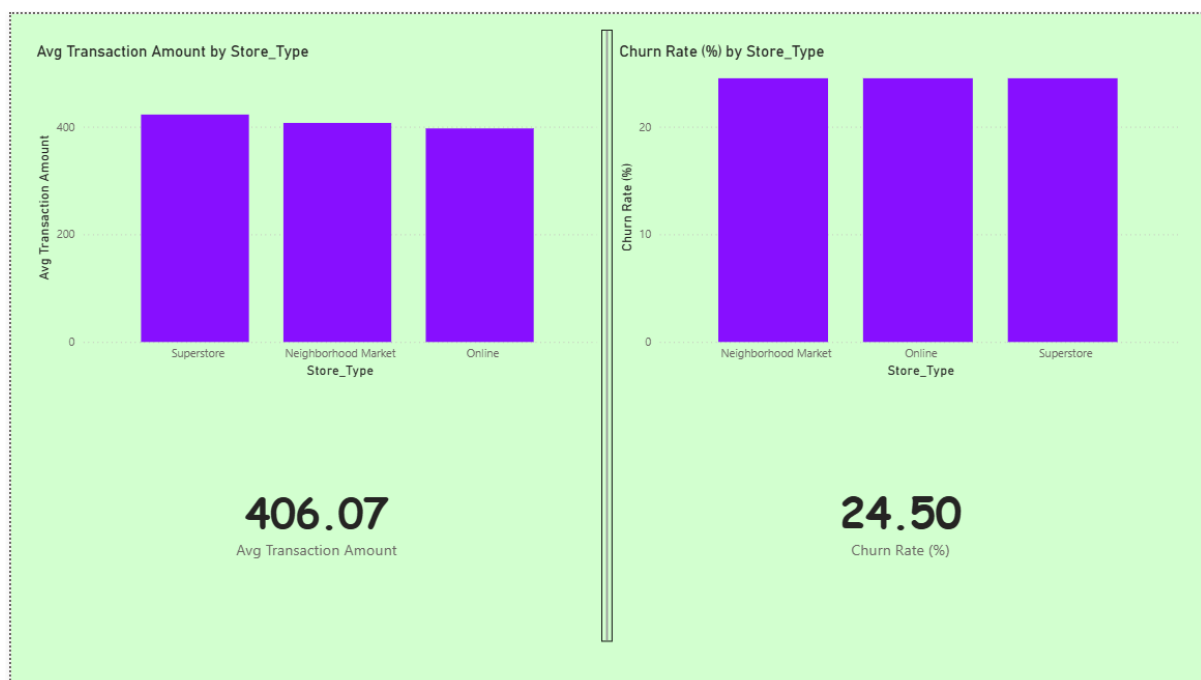
- Give recommendations to improve redemption & retention.
 - Simplify points redemption by lowering minimum limits, allowing partial redemption, and auto-applying points to reduce customer friction.
 - Increase loyalty value through tier-exclusive rewards, early sale access, and experiential benefits.
 - Improve point usage with expiry reminders and personalized nudges.
 - Reduce churn using segment-based retention strategies and personalized offers driven by customer data.
 - Optimize promotions by focusing on high-retention categories and testing non-discount incentives.
 - Strengthen the loyalty program by fast-tracking tier upgrades, rewarding milestones, and gamifying progress to encourage repeat purchases.

Task 5: Store & Channel Performance vs Retention

- Merge store data with transactions

Transaction_ID	Customer_ID	Store_ID	Store_Location.Store_ID	Product_Category	Transaction_Date	Year	Month Name	Amount
1 T0857	C001	S15	S15	Apparel	29-06-2024	2024	June	54
2 T0454	C001	S12	S12	Home & Living	16-03-2023	2023	March	5
3 T0818	C001	S14	S14	Electronics	23-10-2024	2024	October	63
4 T0550	C001	S02	S02	Electronics	02-07-2024	2024	July	76
5 T0531	C001	S20	S20	Apparel	19-10-2024	2024	October	33
6 T0597	C001	S09	S09	Groceries	11-06-2024	2024	June	55
7 T0274	C001	S19	S19	Home & Living	08-11-2023	2023	December	46
8 T0558	C002	S18	S18	Electronics	24-03-2024	2024	March	23
9 T0836	C002	S11	S11	Groceries	18-02-2024	2024	February	36
10 T0580	C002	S02	S02	Apparel	27-05-2023	2023	May	26
11 T0800	C002	S07	S07	Apparel	14-03-2023	2023	March	36
12 T0024	C002	S14	S14	Groceries	25-03-2024	2024	March	10
13 T0805	C003	S17	S17	Groceries	26-10-2023	2023	October	2
14 T0797	C003	S01	S01	Apparel	05-01-2024	2024	January	5
15 T0596	C004	S20	S20	Apparel	21-06-2023	2023	August	32
16 T0776	C004	S06	S06	Groceries	23-07-2024	2024	July	10
17 T0530	C005	S15	S15	Home & Living	21-10-2023	2023	October	67
18 T0758	C005	S09	S09	Electronics	14-08-2024	2024	August	12
19 T0355	C005	S13	S13	Apparel	06-08-2024	2024	August	21
20 T0544	C005	S16	S16	Groceries	29-05-2023	2023	May	60
21 T0023	C005	S09	S09	Groceries	11-12-2023	2023	December	14
22 T0631	C006	S15	S15	Home & Living	06-07-2024	2024	July	39
23 T0706	C006	S02	S02	Home & Living	21-05-2024	2024	May	8
24 T0243	C006	S01	S01	Electronics	06-06-2024	2024	June	30
25 T0202	C006	S01	S01	Apparel	07-08-2024	2024	August	9
26 T0651	C006	S03	S03	Apparel	16-05-2024	2024	May	3
27 T0035	C006	S07	S07	Apparel	21-06-2024	2024	June	5
28 T0883	C007	S10	S10	Apparel	08-08-2023	2023	August	32
29 T0210	C007	S05	S05	Groceries	11-05-2024	2024	May	36
30 T0669	C007	S10	S10	Apparel	16-02-2024	2024	February	5

- Visualize:



```
Avg Transaction Amount =
AVERAGE(Customer_Transactions[Amount])
```

```
Churn Rate (%) =
DIVIDE(
    [Churned Customers],
```

```

    [Total Customers],
    0
) * 100

```



```

Retention Rate (%) =
DIVIDE(
    [Retained Customers],
    [Total Customers (Store)],
    0
) * 100

```

Task 6: Customer Lifetime Value

- $CLV = \text{Total Amount Spent} / \text{Membership Duration (Years)}$

Membership_Duration_year	Customer_ID	CLV
7	C006	107.11
7	C008	120.65
7	C012	258.16
7	C014	463.16
7	C036	284.97
7	C041	113.96
7	C042	147.50
7	C047	429.08
7	C069	291.59
7	C071	70.29
7	C074	346.61
7	C077	426.53
7	C078	128.17
7	C081	403.44
7	C086	240.04
7	C097	343.47
7	C099	388.09
7	C105	210.58
7	C109	287.75
7	C115	366.42
7	C116	157.40
7	C123	111.45
7	C133	71.37
7	C135	438.96
7	C150	165.08
7	C151	253.45
7	C174	237.76
Total		6,863.03

Membership_Duration_year
 7

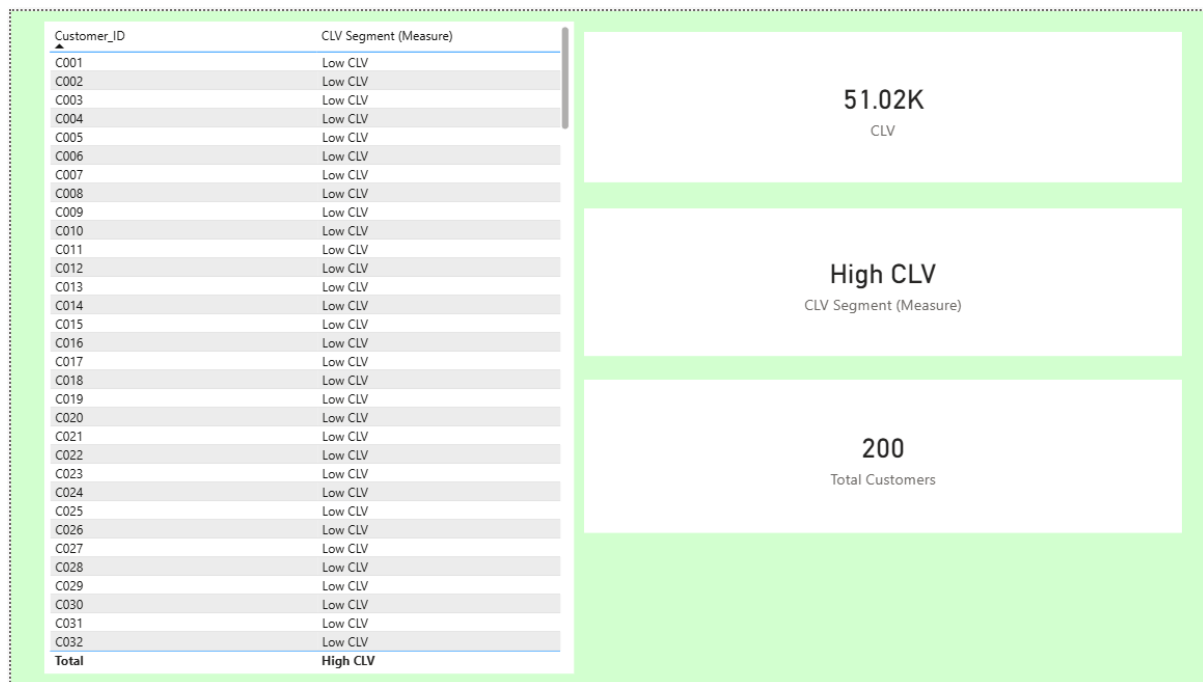
6.86K
 CLV

```

CLV =
DIVIDE(
  [Total Amount Spent],
  [Membership Duration (Years)],
  0
)

```

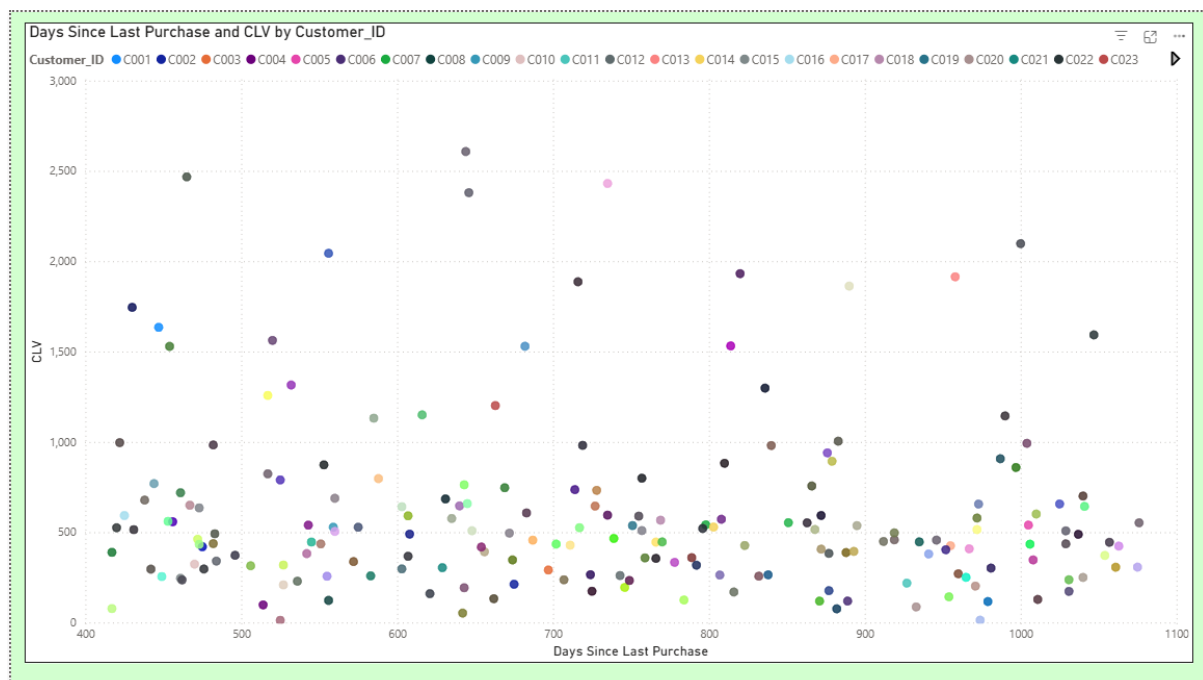
- Segment customers into Low, High CLV
 - Above Average CLV as High
 - Below Average CLV value- Low



- Visualize:

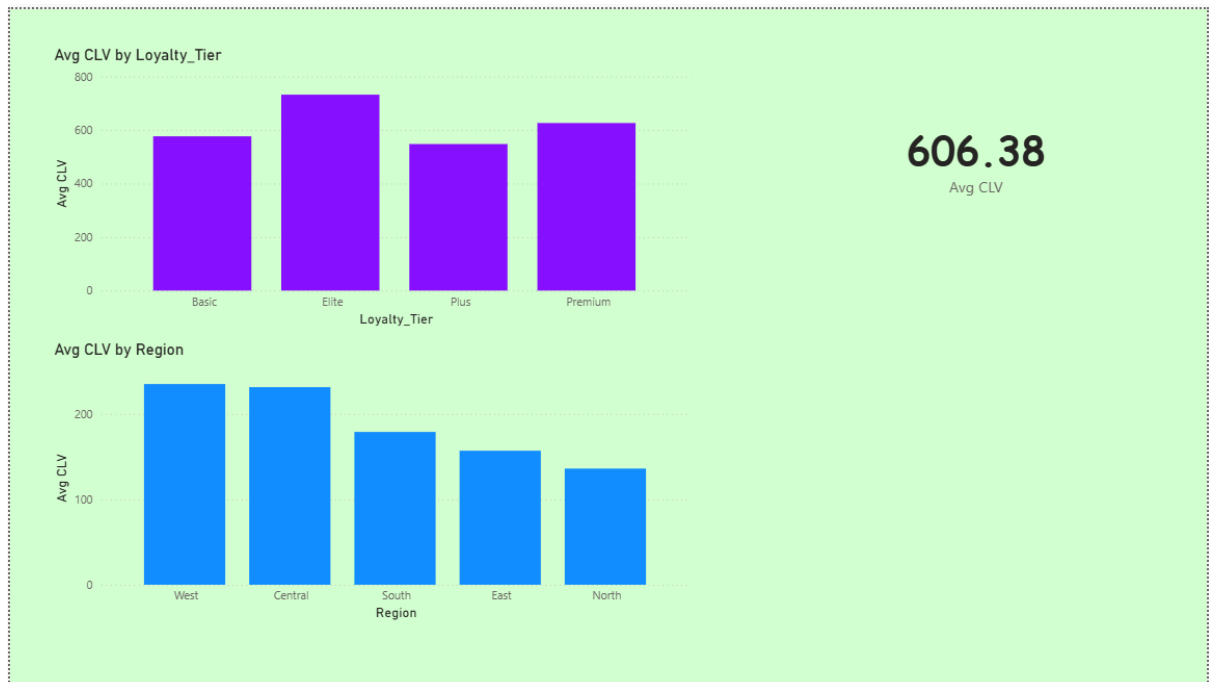
- CLV vs Days Since Last Purchase

- CLV by Loyalty Tier & Region



Days Since Last Purchase =

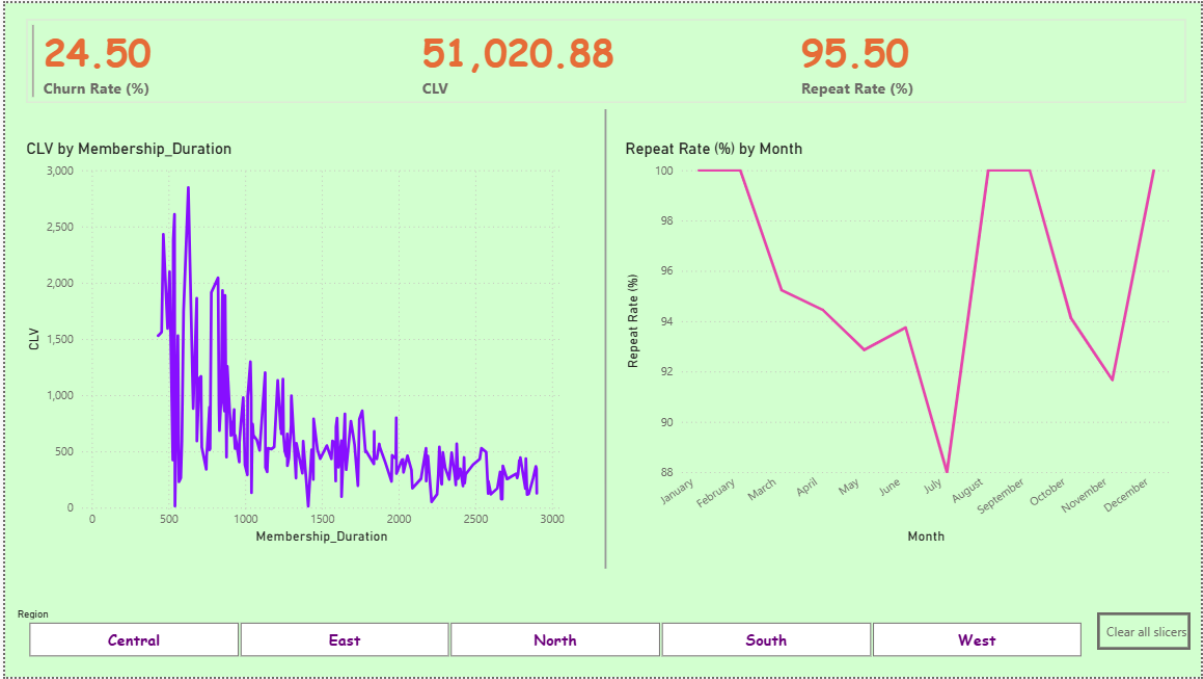
```
DATEDIFF(
    MAX(Churn_Labelled_Customers[Last_Purchase_Date]),
    TODAY(),
    DAY
)
```



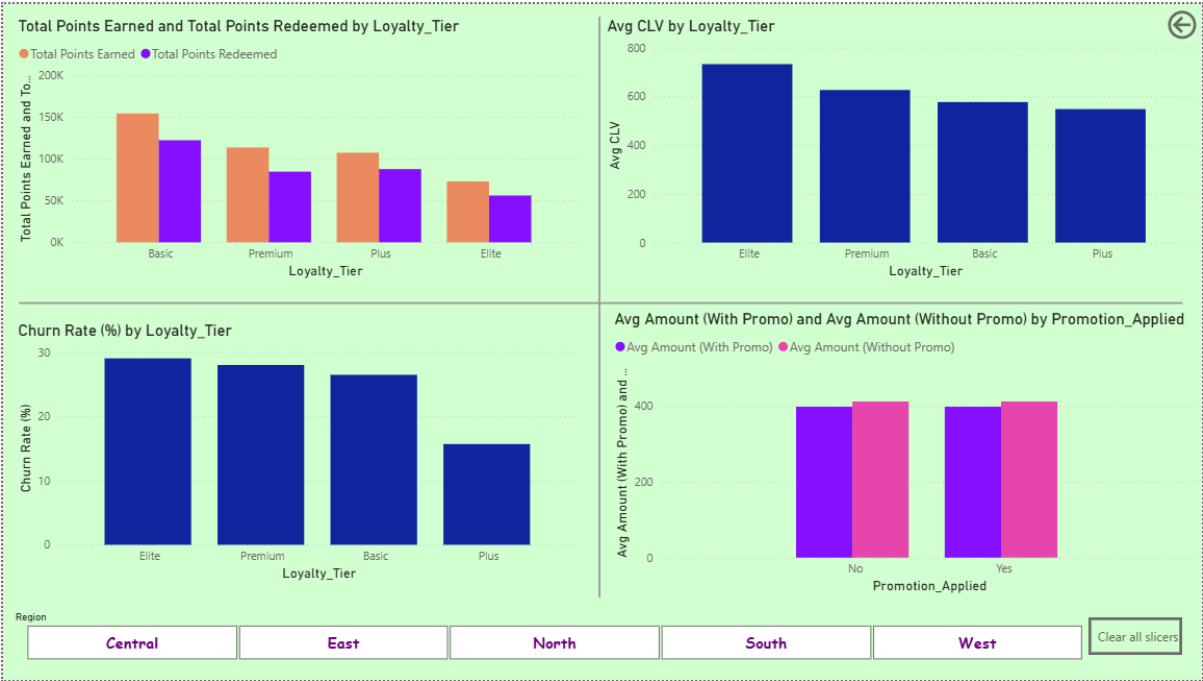
```
CLV =  
DIVIDE(  
    [Total Amount Spent],  
    [Membership Duration (Years)],  
    0  
)
```

Task 7: Final Dashboard & Executive Summary

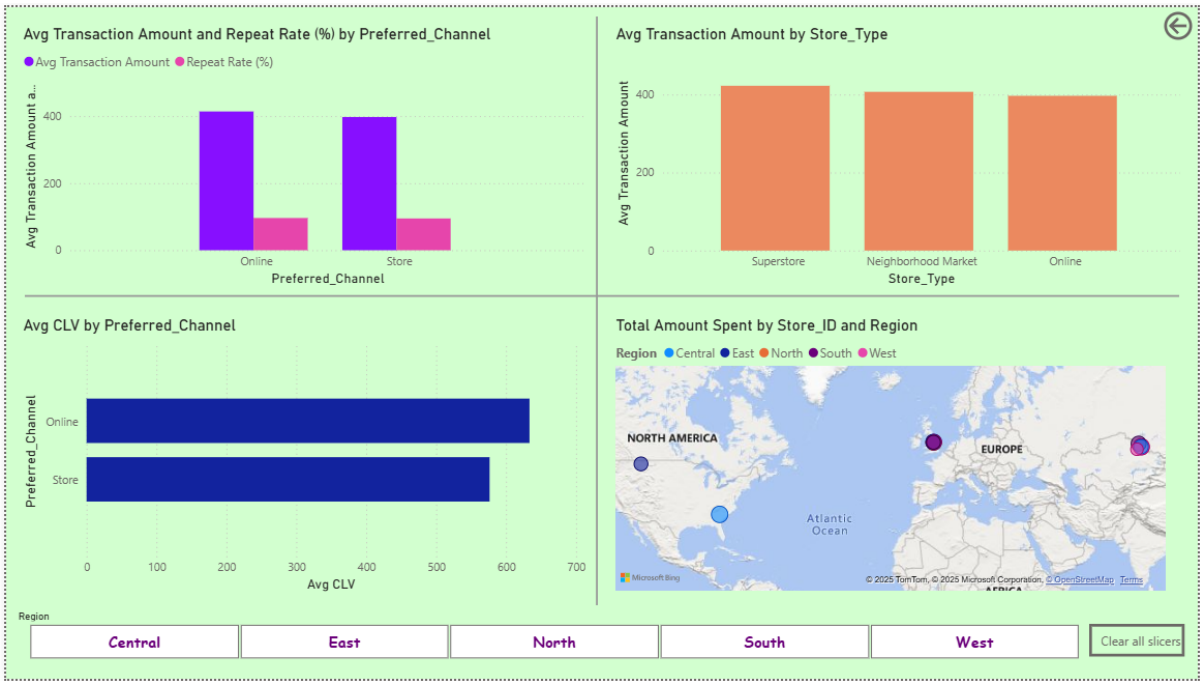
- Multi-page Power BI Report:
- Page 1: KPIs (Churn, CLV, Repeat Rate)



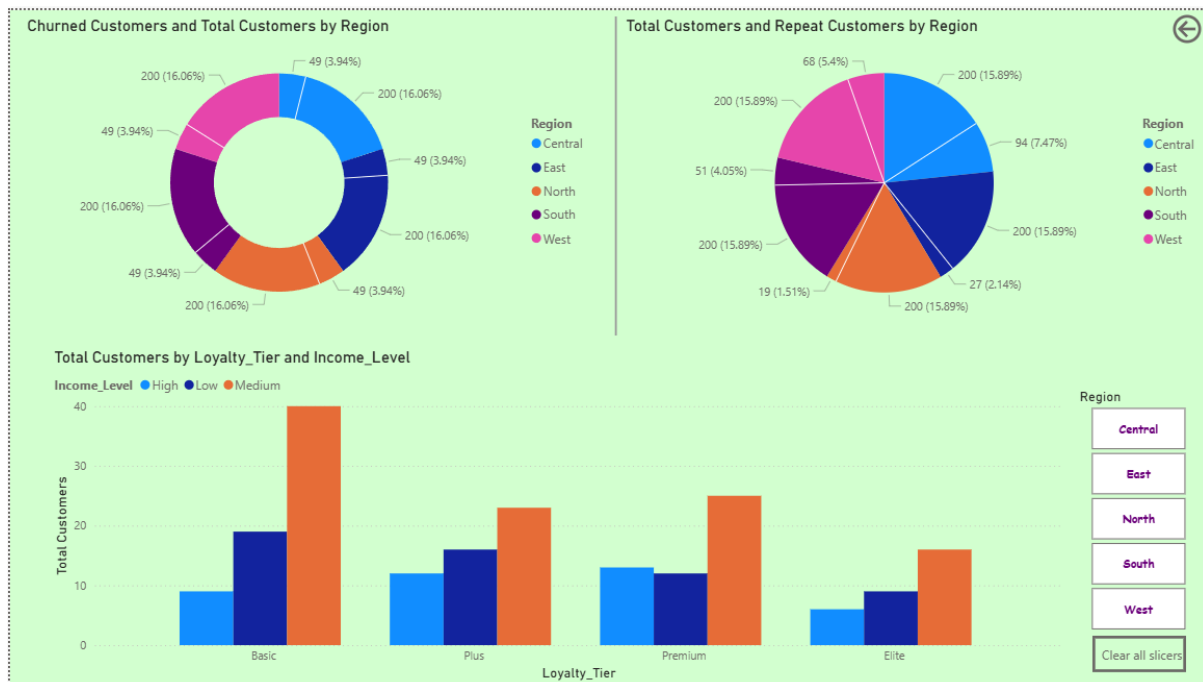
o Page 2: Loyalty & Promotion Impact



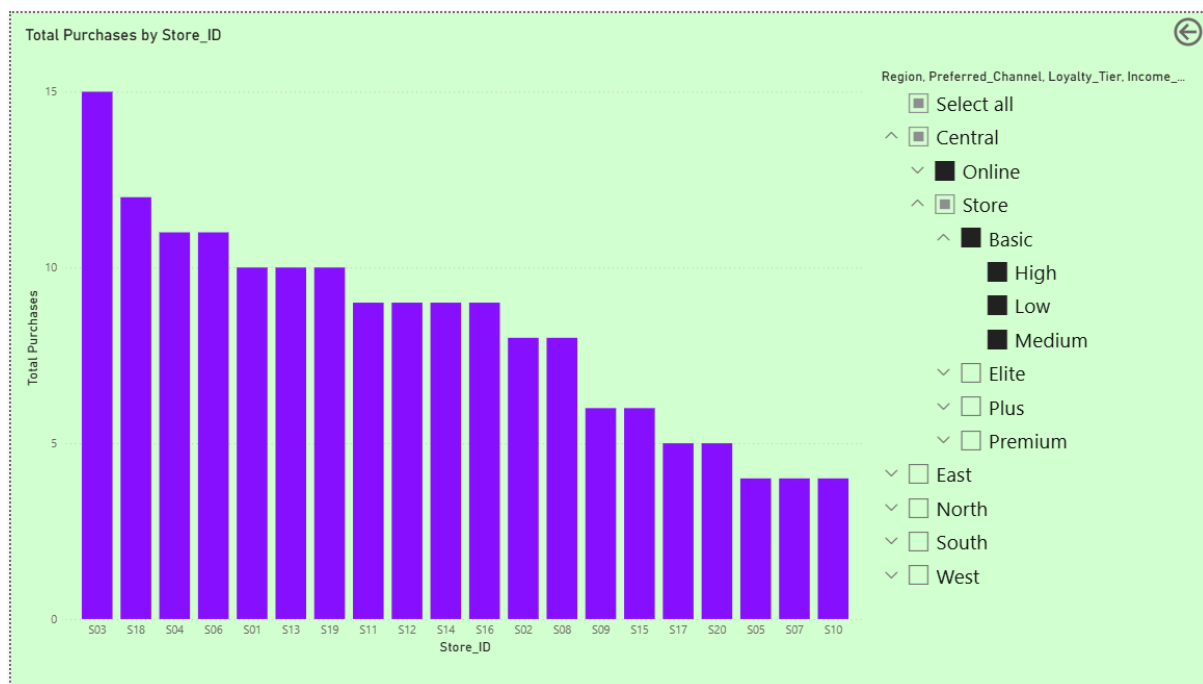
o Page 3: Store/Channel Insights



o Page 4: Segmentation (Churned, Repeat, High-Value)



- Slicers: Region, Channel, Income, Loyalty Tier



- Provide Top 3 Recommendations for Target:

- Which customers to prioritize for retention?

- The High or Above-average CLV ,High = “Days Since Last Purchase”

Mostly in Gold / Platinum loyalty tiers ,because they generate the most revenue per year Losing them causes maximum revenue loss

- Which channels are underperforming?

- Online or Digital Channels are underperforming due to Higher churn rate than offline channels Because of it we get Lower retention rate Customers are often one-time or promo-driven which tends to Lower loyalty engagement and CLV

- How to strengthen loyalty engagement?

- Make loyalty easy to use by lowering redemption limits, allowing partial redemption, and auto-applying points at checkout. Increase perceived value with tier-based exclusive benefits like early access, free delivery, and priority support. Gamify the program using progress bars, milestones, and purchase streak rewards. Use personalized offers and reminders based on customer behavior and inactivity. Ensure loyalty works consistently across all channels (online & in-store). Track key metrics like redemption rate, repeat purchase rate, churn by tier, and CLV uplift.